Displacement and Motion Measurement

Displacement and Motion Measurement

- In the next two lectures we will discuss how to measure various types of displacement and motion
- These include:
  - Linear Displacement
  - Angular Displacement
  - Linear Velocity
  - Angular Velocity
  - Acceleration
  - Force

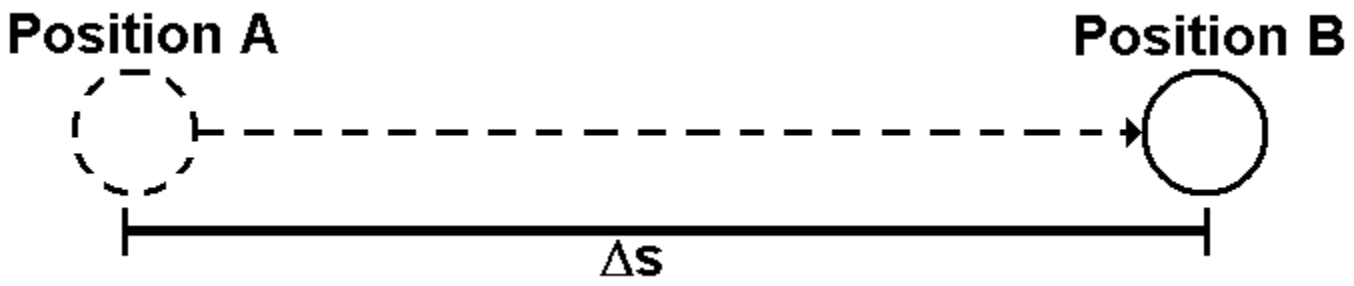
What is Motion?

- Motion or velocity can be defined in terms of the change in position of an object with respect to time
- All measurands of motion are some form of derivative of displacement with respect to time



Linear Displacement

- Is defined as specified distance in a specified direction
- It is measured in length units such as metres, kilometres etc.
- Symbol is s



Measuring Linear Displacement

- Very small displacements:
  - Strain Gauges
  - Capacitive Sensors
  - Inductive Sensors (LVDT)
- Medium displacements
  - Slide Wire / Film
  - Wire wound potentiometer
- Large Displacements (above range of most ‘pure’ linear transducers)
  - Convert linear to angular motion and measure the angular motion with an angular displacement transducer
  - Measure velocity and integrate signal to obtain displacement

Linear Displacement - Resistive Methods

- Resistance is defined by the following equation

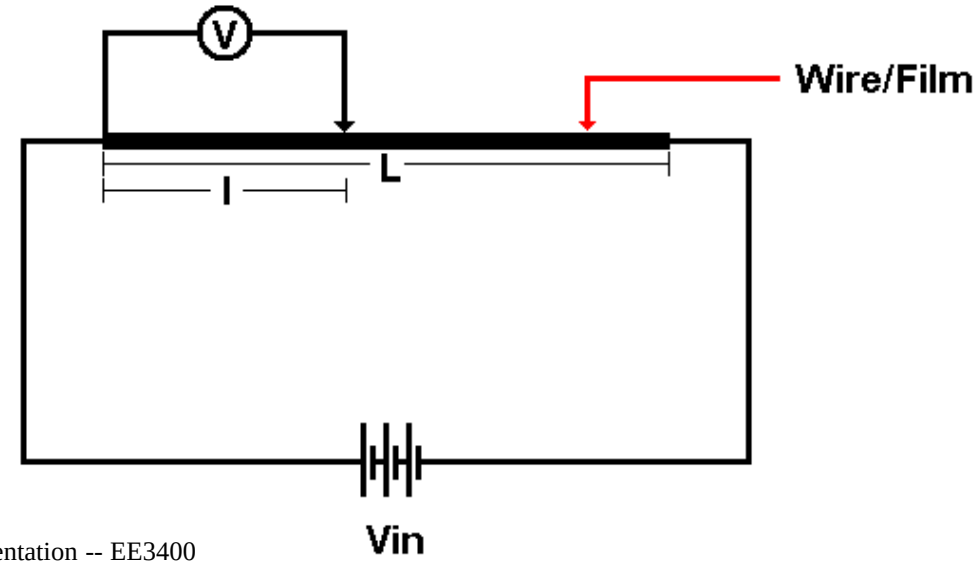
$$R = \frac{\rho l}{A}$$

- Therefore if the length, thickness or resistivity of an object changes with respect to displacement we can use the resistance as a way to measure it

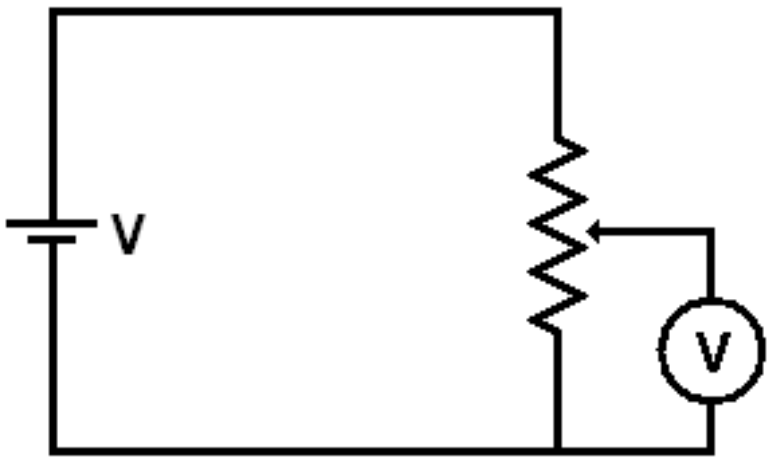


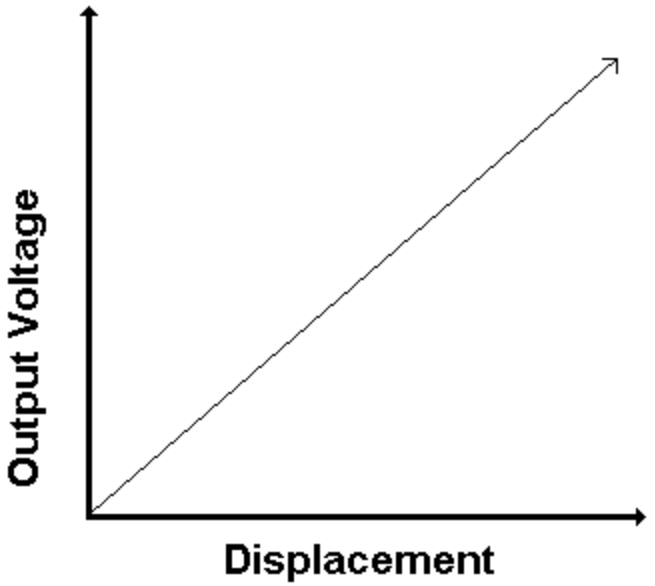
Linear Displacement - Resistive Methods (Slide Wire/Film)

- This is the simplest way of measuring displacement between a moving and a stationary object
- A piece of wire or film is connected to a stationary object
- A slide, which makes contact with the wire, is attached to the moving object
- This acts as a very basic potentiometer



Linear Displacement - Resistive Methods (Slide Wire/Film)



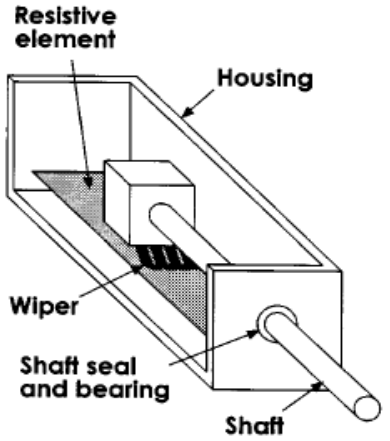


Pros and Cons – Slide Wire

- Range
  - ± 1 – 300mm
- Advantages
  - Simple
  - Good Resolution
  - Low Cost
- Disadvantages
  - Wire does not have high resistance, film is better (±200 to 500Ω/cm)
  - Wear
  - Frictional Loading
  - Inertial Loading

Linear Displacement - Resistive Methods (Wire Wound Potentiometer)

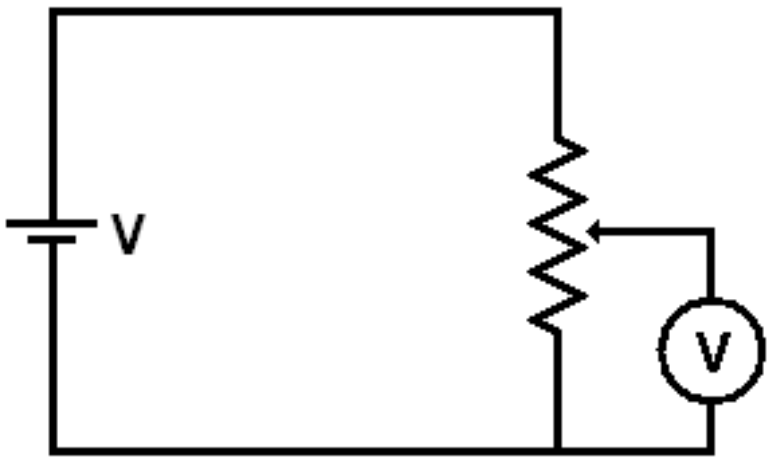
- Wire Wound potentiometers use the same principle as slide wire sensors except that they use a coil of insulated resistance
- The slider runs on one surface of the coil that is not insulated

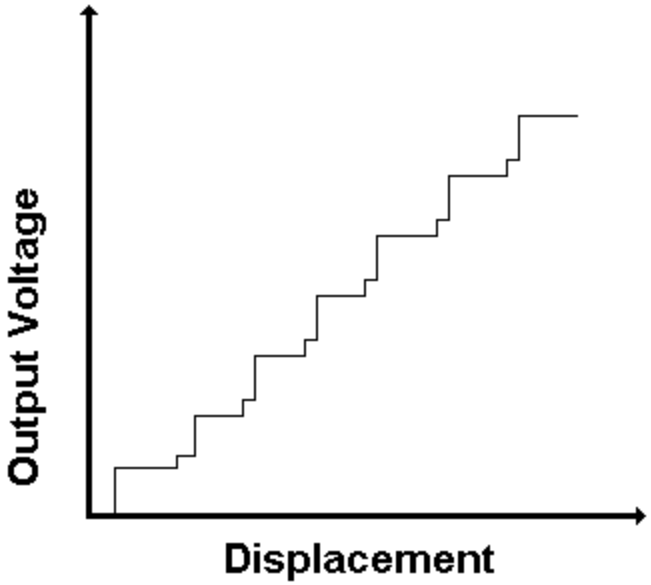




Taken from Novotechnik

Linear Displacement - Resistive Methods (Potentiometer)





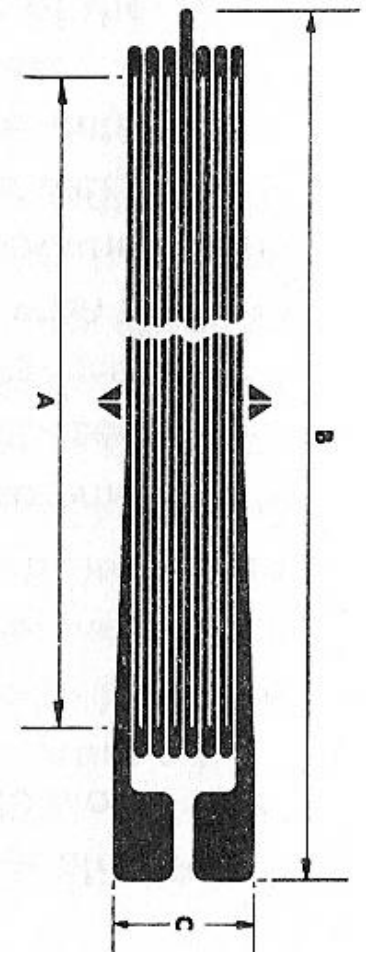
Pros and Cons – Potentiometers

- Resolution
  - ± 1mm – 4m
- Advantages
  - Simple
  - Robust
- Disadvantages
  - Resolution dependant on wire diameter
  - Continuous use over portion of the wire will cause wear
  - Frictional Loading
  - Inertial Loading

Linear Displacement - Resistive Methods (Strain Gauges)

- Attach the strain gauge to the object
- When the object is in tension or compressed it will result in a change in the resistance of the strain gauge
- This is used to measure the change in length of the object

$$\frac{\Delta R}{R} = S\frac{\Delta L}{L}$$



Pros and Cons – Strain Gauges

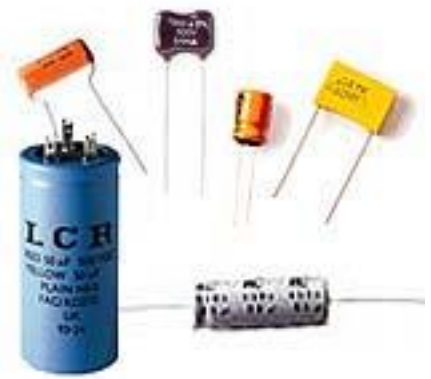
- Advantages:
  - Relatively easy to understand and attach
  - Cheap
- Disadvantages
  - Need temperature compensation

Linear Displacement - Capacitive Methods

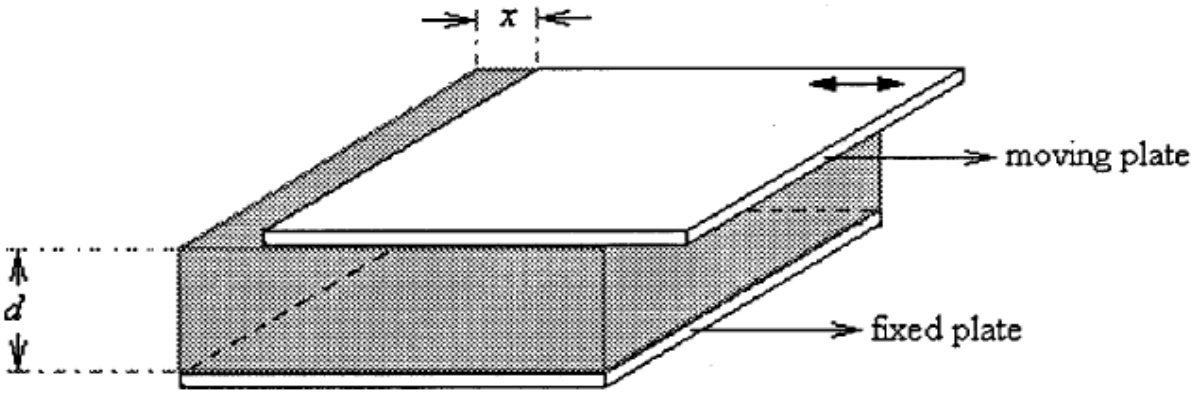
- As we have seen in a previous lecture capacitance is defined as
- Therefore we could use the change in
  - Plate Area
  - Permittivity of the dielectric
  - Distance between the plates

  as a way to measure displacement

$$C = \frac{A\varepsilon_0\varepsilon_r}{d}$$



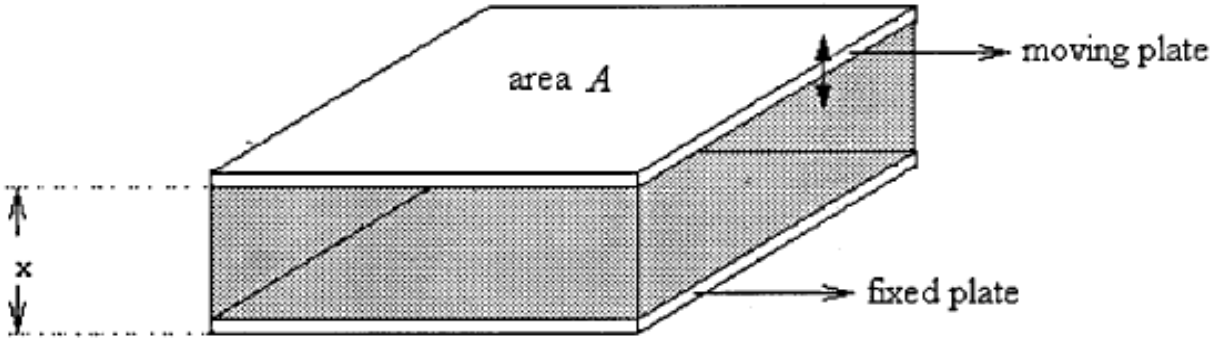
Linear Displacement - Capacitive Methods (Variable Area)



- If we have an two electrodes and one moves relative to the other in a linear direction we will get an effective change in the area of the plates
- This results in a change in the capacitance which can be related to displacement

$$C = \frac{(Aw)\varepsilon_0\varepsilon_r}{d}$$

Linear Displacement - Capacitive Methods (Distance Between the Plates)

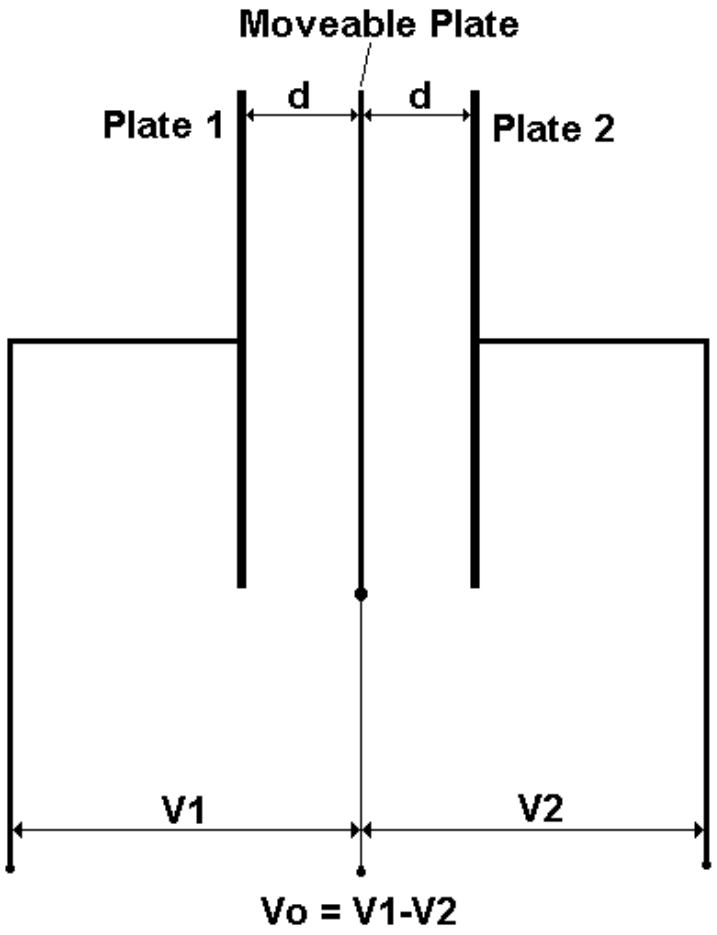


- If we have to electrodes, one fixed and the other movable we can arrange it that the distance between the plates changes for a change in displacement

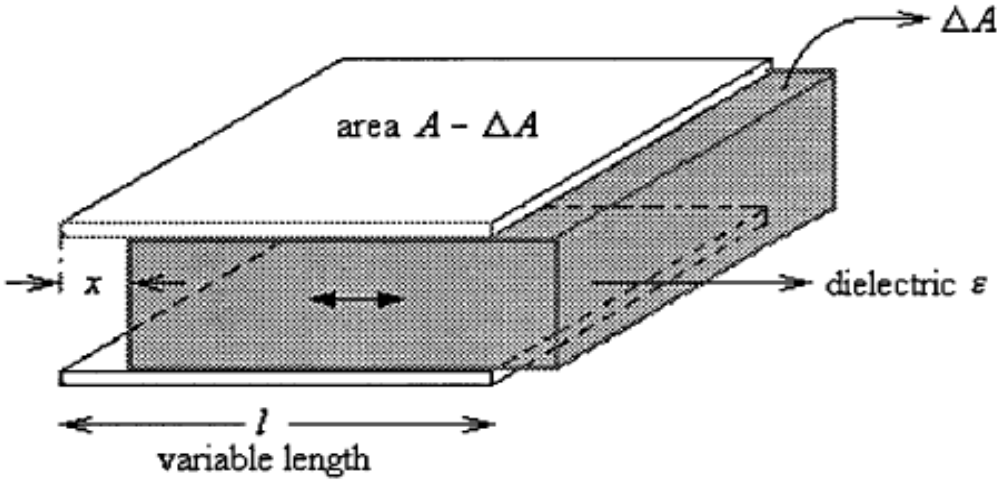
$$C = \frac{A\varepsilon_0\varepsilon_r}{x}$$

Linear Displacement - Capacitive Methods (Distance Between the Plates)

- This type of capacitive arrangement consists of two fixed outer plates and one central moveable plate
- The central plate can move towards either of the plates which essentially changes the capacitance between the moveable plate and the fixed plates
- If the moveable plate is in the centre of the capacitor, voltages V1 and V2 will be equal



Linear Displacement - Capacitive Methods (Permittivity)



- The dielectric moves relative to the plates and this causes a change in the relative permittivity of the dielectric

Linear Displacement - Inductive Methods

- Inductive methods use very similar principles to resistive and capacitive methods
- The inductance of a coil is given by the following equation

$$L = \frac{N^2 \mu A}{l} \quad [H]$$

- Where N is the number of turns in the coil, μ is the effective permeability of the medium in and around the coil, A is the cross sectional area and l is the length of the coil in m
- As with the other examples if we change any one of these parameters we get a change in the inductance

Linear Displacement - Inductive Methods
(Linear Variable Differential Transformers LVDTs)

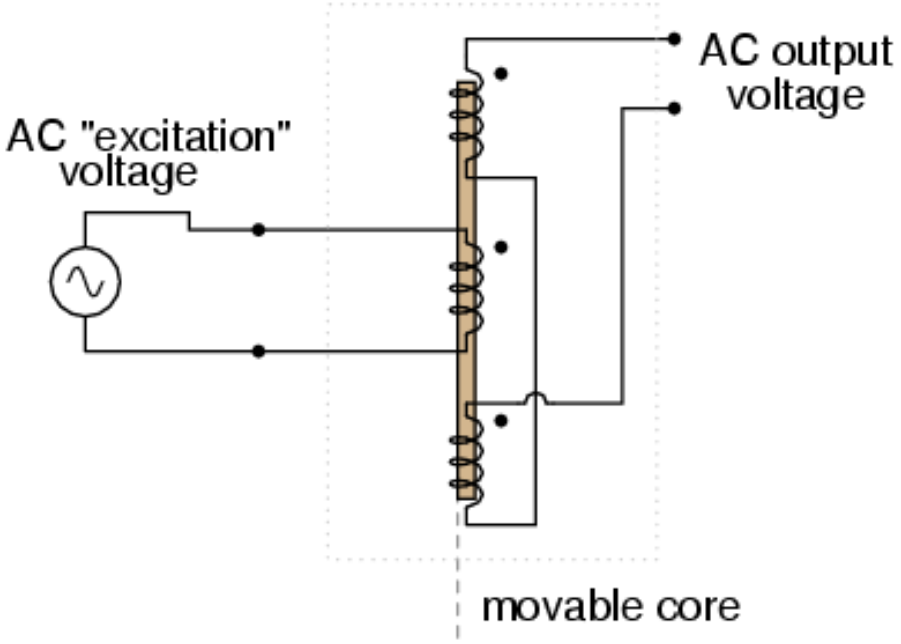
- LVDTs are accurate transducers which are often used in industrial and scientific applications to measure very small displacements



Linear Displacement - Inductive Methods

(Linear Variable Differential Transformers LVDTs)

- An LVDT consists of a central primary coil wound over the whole length of the transducer and two outer secondary coils
- A magnetic core is able to move freely through the coil



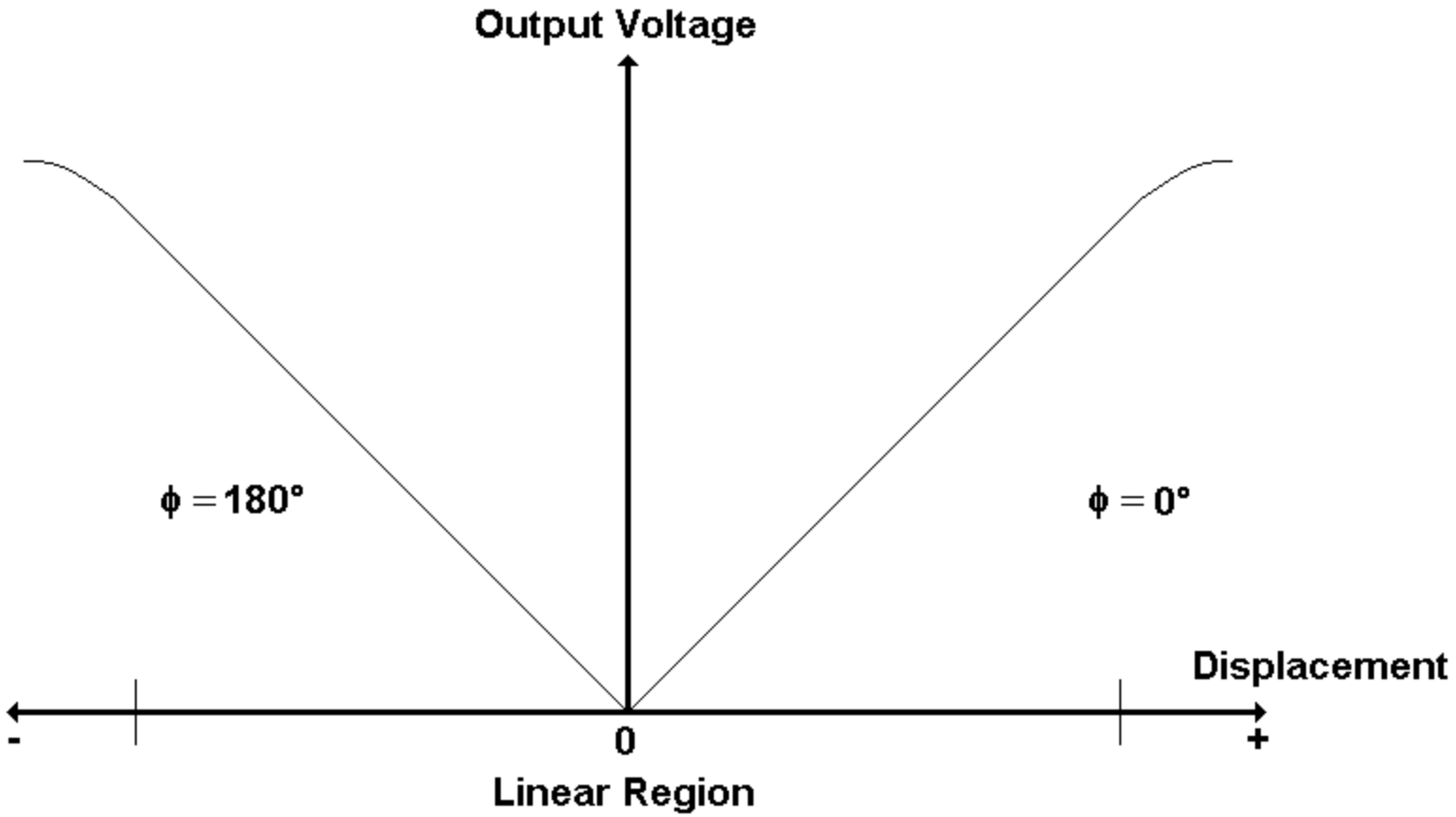
Taken from www.allaboutcircuits.com

Linear Displacement - Inductive Methods (Linear Variable Differential Transformers LVDTs)

- The primary windings are energized with a constant amplitude AC signal (1 – 10kHz)
- This produces an alternating magnetic field which induces a signal into the secondary windings
- The strength of the signal is dependant on the position of the core in the coils
- When the core is placed in the centre of the coil the output will be zero
- Moving the coil in either direction causes the signal to increase
- The output signal is proportional to the displacement

Linear Displacement - Inductive Methods

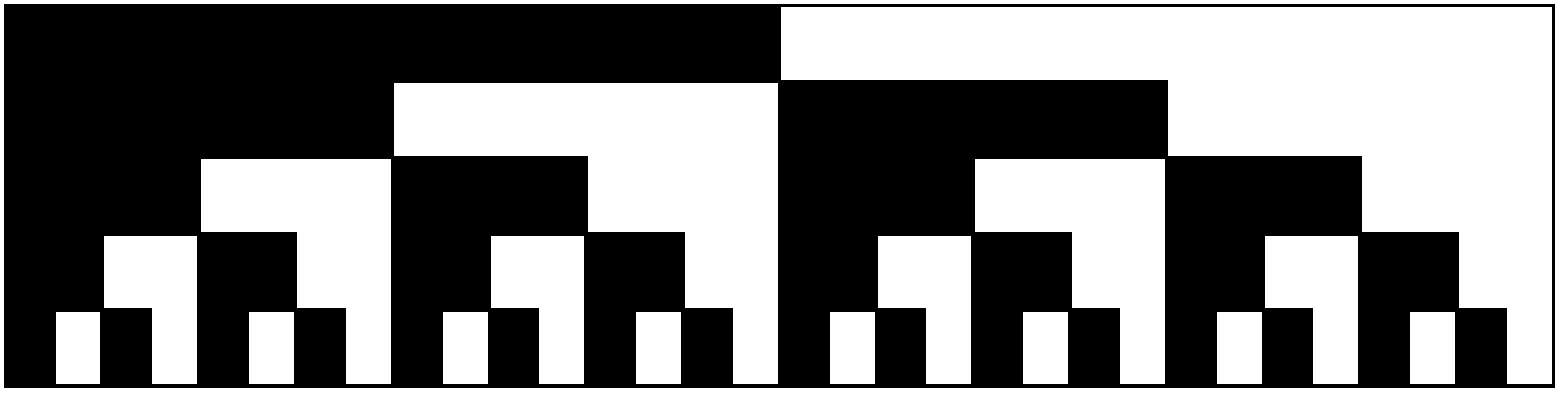
(Linear Variable Differential Transformers LVDTs)



Pros and Cons - LVDTs

- Range: ±2.5nm - ±10cm
- Advantages:
  - Good resolution
- Disadvantages:
  - Needs shielding to prevent interference from magnetic sources

Linear Displacement – Digital Methods



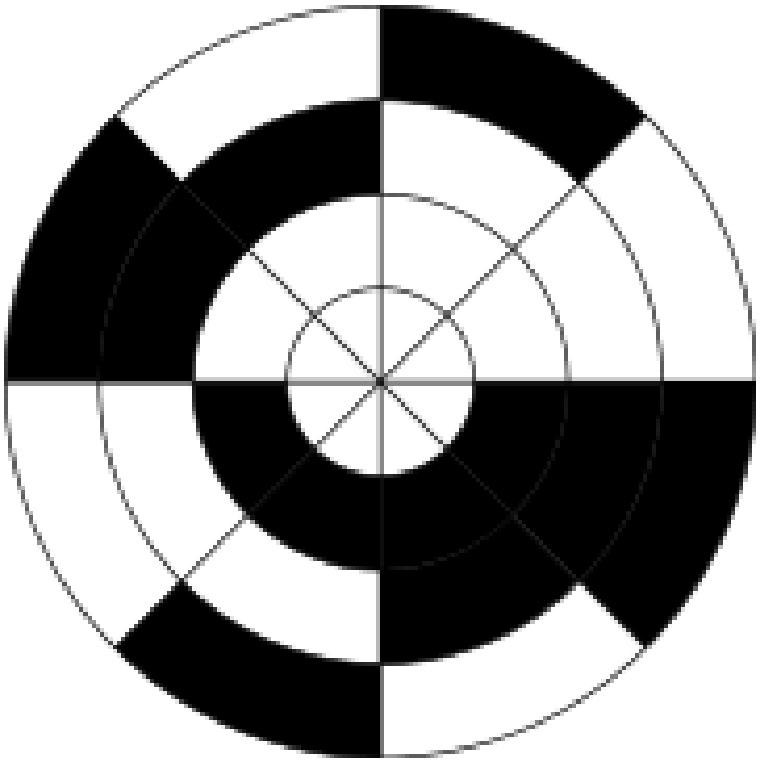
- It is also possible to measure displacement using digital methods
- As a binary system only uses 0's and 1's we can represent this using transparent and opaque areas on a glass scale or conducting and non conducting areas on a metal scale
- Each position will produce a unique code which represents a specific displacement

Angular Displacement

- Angular displacement (θ) of an object is the angle in radians through which a point has been rotated about an axis
- Many of the transducers used to measure linear displacement have counterpart which measure angular displacement

Angular Displacement – Digital Methods

- Known as a rotary encoder
- Operates on a similar principle to the digital methods used to measure linear displacement
- Typically uses a light source and detector to detect
  - Reflective and non reflective areas or
  - Transparent and opaque areas
- Often used in robotics



Linear Velocity

- Is the rate of displacement of an object or the speed of an object in a particular direction
- Linear velocity can be found by:
  - Differentiating displacement
  - Integrating acceleration
- Typically an integration operation is perform as it also acts as a low pass filter which will improve the signal to noise ratio of the device

$$v = \frac{ds}{dt}$$

Linear Velocity - Digital Methods

- Have a number of evenly spaced marks on the moving object
- A detector monitors how many marks pass a certain point in a given time



$$v = \left(\frac{nd}{t}\right) \quad [ms^{-1}]$$

Linear Velocity - Doppler Effect Methods

- This is the method typically used in police radars
- It is a very effective and accurate methods
- A narrow radio or ultrasonic beam is aimed at an object
- This will be reflected back to the source
- If the object is moving the frequency of the received signal will be different to the transmitted signal
- The difference between these two frequencies is a measure of the linear velocity of the moving object
- The received frequency will be:
  - Higher if the object is moving towards the source
  - Lower if the object is moving away from the source

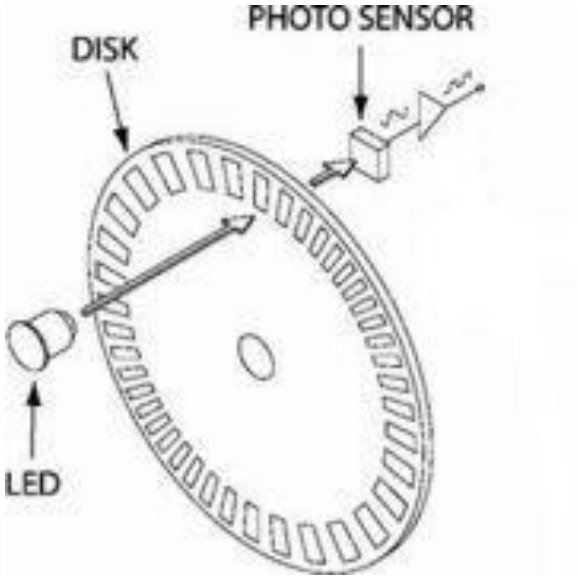
Angular Velocity

- Very common devices
- Tachometers are used in motor control applications, are found in most cars etc.
- Often used to measure linear velocity for example a car speedometer



Angular Velocity - Digital Methods

- Typically uses evenly spaced marks around a shaft or as evenly space holes in a disc
- A pulse train will be produced and its frequency will be proportional to the angular velocity of the system



Acceleration

- Is the rate of increase in velocity
- The SI unit is m/s^2
- It is also the second derivative of displacement
- Linear acceleration is measured with accelerometers



Accelerometers

- Used to measure acceleration or gravitational force
- Single or multi axis accelerometers are available
- Used to measure:
  - Acceleration
  - Vibration
  - Tilt
  - Shock
- Often found in portable devices such as digital cameras







Types of Accelerometer

- There are many different types of accelerometer available which use various methods to detect acceleration
- Here are a few types:
  - Piezoelectric accelerometers
  - Microelectromechanical (MEMs) accelerometers